System Design Interviews

Framework and back-of-envelope maths

The 45-minute framework

- Clarify the scope and the users.
- Functional and non-functional requirements.
- Rough estimates.
- Entities and API.
- High-level diagram.
- Deep dives on bottlenecks.
- Trade-offs and failure modes.

Back-of-envelope numbers

- 1 day ≈ 86,400 s ≈ 10^5 s; 1M requests/day ≈ 12 QPS; peak ≈ 2–3× the average.
- Character 1 B; tweet-size text ~300 B; photo ~200 KB–2 MB; 1 min of HD video ~50–100 MB.
- One server handles ~10k concurrent connections and ~1k–10k simple QPS; Redis does ~100k ops/s.
- A disk sequential read is ~100× faster than random; reading 1 MB from memory takes ~250 µs, from SSD ~1 ms.

Worked estimate: photo sharing

- 100M DAU × 2 uploads/day = 200M photos/day
- 200M ÷ 86,400 s ≈ 2.3k uploads/s (peak ×3 ≈ 7k/s)
- × 2 MB ≈ 400 TB/day → ~150 PB/year (× 3 replicas)
- Reads 50× writes ≈ 115k/s → CDN + cache
- Conclusion: blob store + CDN; metadata DB sharded

Recap

- Same structure every time
- Requirements before boxes
- Estimate, then size the design
- Finish with trade-offs